

India Import Dependency — Sectoral Overview (FY 2024-25)

Executive Summary

India's merchandise import bill reached **\$678.7 billion in FY 2024-25**, generating a trade deficit of **\$241.3 billion**. Ten sectors drive the bulk of this dependency. The structural danger is not the aggregate — it is **China's chokehold on industrial inputs**: imports from China hit **\$113.5 billion in FY 2024-25**, but in individual sectors the exposure is far more acute — 73.7% of pharmaceutical APIs, 88% of integrated circuits, 94% of lithium-ion batteries, and 65% of solar modules are China-sourced. Government countermeasures (PLI, Make in India) have delivered at the assembly level but not at the core industrial level.^{[1][2][3][4][^5]}

Sector Snapshot

Sector	Import Value (USD Bn)	% of Total	Top Supplier	China Share	Risk
Energy & Petroleum	~220.6	~32.5%	Russia (35-36%) [^6]	<2%	HIGH
Electronics & Semiconductors	~89.8	~13.2%	China (~44%) [^3]	44%	VERY HIGH
Precious Metals (Gold/Silver)	~60.5	~8.9%	Switzerland (~40%) [^7]	<3%	MEDIUM
Industrial Machinery	~56.2	~8.3%	China (~30%) [^8]	30%	HIGH
Chemicals & Pharma APIs	~28.4	~4.2%	China (~73.7%) [^4]	73.7%	VERY HIGH
Edible Oils & Fertilizers	~26.8	~3.9%	Indonesia / Russia [^9]	~10%	HIGH
Automotive & EV Chain	~18.5	~2.7%	China (~40%) [^10]	40%	HIGH
Renewable Energy Components	~6.2	~0.9%	China (~65%) [^5]	65%	VERY HIGH

Defense & Aerospace	~5.5	~0.8%	Russia (~36%) [¹¹]	<1%	VERY HIGH
Critical Minerals & Metals	~4.1	~0.6%	Australia / DRC[¹²]	~23%	CRITICAL

Key Structural Risks

China Concentration. India's trade deficit with China widened to a record **\$99.2 billion in FY 2024-25** and further to **\$116 billion in 2025**. China supplies inputs for which no near-term alternative exists at scale — antibiotics (erythromycin 97.7% Chinese-sourced), silicon wafers (96.8%), and semiconductor manufacturing machinery (99.5%).[³¹][⁸][¹³]

Critical Minerals Gap. India imports **100% of its lithium, cobalt, and nickel** — the foundational materials for EVs and energy storage. Domestic mines have zero commercial production, and new projects carry 10-15 year lead times. China controls global midstream refining of these materials, meaning geographic diversification of mines alone does not reduce China exposure.[¹⁴]

Energy Vulnerability. ~50% of crude oil transits the Strait of Hormuz. Strategic petroleum reserves cover only **74 days of net imports**. Russia's emerging sanctioned status post-2025 adds further supply unpredictability to India's largest crude supplier.[⁶¹][¹⁵]

Food & Agriculture. India is the world's largest edible oil importer at ~60% import dependence. China's temporary fertilizer export halt in 2025 directly disrupted Kharif sowing — urea imports surged **136.6% YoY** — illustrating how a single supplier decision translates into food security risk.[¹⁶¹][¹⁷]

PLI & Make in India — What's Working, What Isn't

The PLI scheme has generated ₹1.76 lakh crore in investments and ₹12.5 lakh crore in incremental production. Successes are real but narrowly concentrated:[¹⁸¹][¹⁹]

- **Mobile phones:** Production rose 28-fold; India surpassed China as the top smartphone exporter to the US in Q3 2025[²⁰]
- **Telecom equipment:** 60% import substitution achieved[¹⁸][¹⁸]

- **Solar modules:** China's share fell from 90% to 65%^{[5][21]}

Persistent failures: India still cannot manufacture semiconductor wafers, battery cells at commercial scale, jet engines, or bulk APIs at competitive cost. PLI has been most effective where assembly was the missing link — it has not yet cracked sectors requiring deep R&D, expensive capex, or technology that China/OECD nations guard closely.

Import Substitution Priority Matrix

Sector	Substitution Potential	Timeline	Key Government Lever
Electronics Components	Very High	5-10 yrs	India Semiconductor Mission, ECMS ^[^20]
Pharma APIs	High	3-7 yrs	PLI Bulk Drugs, 3 Drug Parks ^[^22]
Solar (wafer/ingot)	High	2-5 yrs	PLI Solar PV (extended to 2030) ^[^23]
EV Batteries	High	5-10 yrs	PLI ACC (₹18,100 Cr) ^[^24]
Critical Minerals	Critical	10-15 yrs	NCMM 2025, KABIL overseas mining ^[^24]
Edible Oils	Moderate	7-10 yrs	₹10,103 Cr Oilpalm Mission ^[^25]
Defense Platforms	Moderate	7-15 yrs	75% domestic procurement mandate ^[^11]

Strategic Recommendations

1. **Emergency API buffer stocks** — mandate 6-month reserves for top 53 PLI-targeted APIs; operationalize all three Bulk Drug Parks by 2027.
2. **Extend PLI Solar to wafers/ingots** — current localisation stops at module assembly; upstream capacity is required to truly de-risk China exposure.^[^5]
3. **Fast-track KABIL mineral contracts** — secure long-term offtake agreements in Australia, Argentina, and DRC for lithium and cobalt before global demand tightens post-2027.

4. **Battery cell technology transfer** — structure JVs with Korean or Japanese cell manufacturers (LG Energy, Panasonic) to accelerate ACC PLI outcomes rather than waiting for domestic R&D to catch up.
5. **Strategic petroleum reserve expansion** — raise SPR capacity to 90-day net import equivalency and institutionalise futures hedging for PSU refiners.

Sources: Ministry of Commerce & Industry^{[26][1]}, PIB^{[4][18][^19]}, SIPRI Arms Report 2024^{[27][28]}, ICRA^{[29][17]}, IEEFA^[^14], GTRIC/GTAI bilateral trade analysis^{[3][8]}, Reuters^[^25], Newsbase/AngelOne^{[5][23]}, Economic Times^{[8][13]} | FY 2024-25 data unless noted.

References

1. In-Depth Analysis of India's Export-Import Data for FY 2024-25 - Total Exports: \$820.93B, Imports: \$915.19B, Deficit: -\$94.26B. The table reveals that services expor...
2. Export and Import Performance in Financial Year 2024-2025. - Total Imports (April-December 2024): Estimated at USD 677.44 billion, a 5.66% decrease from USD 718....
3. India-China Trade Dependencies: Doubling Flows ... - GTAIC - India's imports from China have nearly doubled within eight years, rising from \$71.97 billion in 201...
4. India's dependence on imported APIs - PIB - In the Financial Year 2024-25, India imported 200 categories of APIs, bulk drugs, and drug intermedi...
5. India Cuts Solar Imports: Dependence on China Drops to 56% for ... - India's solar cell and module imports fell by 20% and 57% in the first 8 months of 2024-25, driven b...
6. Russian crude never left India's import mix - ThePrint - Russian crude never left India's import mix. It made up 1/3rd of oil imports from 2024 to 2026. Afte...

7. [Gold imports jump in January, Swiss, UAE inflows lead spike](#) - Gold imports increased from \$32.9 billion in 2018-19 to \$58 billion in 2024-25, a 76 percent increas...
8. [India-China trade gap: Deficit widens to \\$99.2 billion in 2024-25](#) - The trade deficit —calculated as the difference between imports and exports—has expanded from \$1.1 bi...
9. [India's edible oil imports remain flat in oil year 2024-25, import bill ...](#) - While total palm oil imports declined to 75.82 lt in oil year 2024 ... During the oil year 2024-25, ...
10. [India's EV Supply Chain & Value Addition - LinkedIn](#) - Battery costs account for 30-40% of the electric vehicles cost and India is dependent on imported ce...
11. [India 2nd largest Arms Importer: SIPRI - Drishti IAS](#) - India's share of global arms imports fell to 8.3% in 2020-24, making it the 2nd-largest arms importe...
12. [\[PDF\] India's Hunt for Critical Minerals - IEEFA](#) - 100% import dependency for minerals like lithium, cobalt and nickel. ... Minerals such as lithium, n...
13. [Indian exports to China rise in 2025 but trade deficit touches record ...](#) - India's exports to China saw a significant rise in 2025, climbing to USD 19.75 billion. This growth ...
14. [India's critical minerals push faces funding gap: IEEFA - Argus Media](#) - India imports 100pc of the lithium, cobalt and nickel used in clean energy manufacturing, with deman...
15. [India increases Russian crude imports by 45% amid Middle East ...](#) - India's crude imports from Russia averaged 1.5 million barrels per day on March 1-12, up 41% from Ja...
16. [India Fertiliser Imports Rise Amid Domestic Production Decline](#) - "India imported 136.6% more urea, as the domestic production of the most used fertilisers dropped 4%...
17. [\[PDF\] INDIAN EDIBLE OIL INDUSTRY - ICRA](#) - About ~60% of total edible oil consumption is imported; palm oil constitutes ~40% of consumption whi...

18. PLI scheme incentivizes domestic manufacturing, increases ... - PIB - India has achieved 60% import substitution in telecom products under the PLI scheme for Telecom & Ne...
19. Government Scales Up PLI Budget to Accelerate Manufacturing - PIB - India has achieved 60% import substitution in telecom products under the PLI scheme. Global tech com...
20. India is dependent on China for electronic components. Now it's ... - The country exported electronics worth \$38.56 in 2024-25, while importing electronic components worth...
21. China's Share in India's Solar Cell Import • FY24 : 90% • FY25 : 56 ... - China's Share in India's Solar Cell Import • FY24 : 90% • FY25 : 56% PLI Scheme Uplifted Local Manuf...
22. Production Linked Incentives (PLI) of India - Reviewing the Impact - The overall aim of the scheme is to strengthen the manufacturing sector and reduce dependence on im...
23. India moves to cut reliance on Chinese solar modules and PV cells - In value terms, solar module imports stood at \$4.35bn in FY2024, recording 360% growth compared to F...
24. impact on ev due to policy changes in china pertaining to lithium - PIB - The Government has taken various steps to reduce import dependence for critical minerals. The Union ...
25. India aims to double edible oil output to trim reliance on imports - The world's largest importer of edible oils, India currently fulfils nearly two-thirds of its demand...
26. The cumulative exports (merchandise & services) during FY 2024 ... - Total imports during FY 2024-25 (April-March)* is estimated at US\$ 915.19 Billion registering a grow...
27. [PDF] Trends in International Arms Transfers, 2024 - SIPRI - The top five arms importers—Ukraine, India, Qatar,. Saudi Arabia and Pakistan— received 35 per cent ...

28. Top 10 largest arms importing countries (2020–2024) - As of the period between 2020 and 2024, India ranked as the world's second-largest arms importer, ac...

29. [PDF] India's API strength endures, China reliance persists - ICRA - The Indian active pharmaceutical ingredients. (API) industry is estimated to be worth about \$13.5 bi...